CARLOS SIMÕES

AI ENGINEERING LEAD | AI ARCHITECT

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EXPERIENCE

Lead AI
Unit4 | Coimbra, Portugal (Remote) | Nov 2025 – Present

- Led AI discovery workshops with directors and department leaders across Finance, Sales Ops, People and Customer Success: mapped 21 opportunities and prioritised 10 through effort estimates and business cases (AI, automation or process redesign).
- Finance (sole engineer): shipped 3 document-AI solutions to production on Azure (audit template completion, expense matching, contract validation against CRM and enterprise databases) on one reusable OCR, LLM extraction and review foundation; est. 720 h/year saved.
- Led and architected 7 use cases live in production with a team of 10 consultants (3 direct reports) across Sales Ops, People and Customer Success, incl. sales rule validation, payroll reconciliation in 22 countries and a Snowflake insight agent; est. 7,800+ h/year saved.
- Platform & standards: designed the architecture and implemented the infrastructure of the shared AI platform on Azure from scratch; every solution traced in Langfuse, with Prometheus and Grafana monitoring being added, for accuracy, answer quality, errors, latency and cost; code, architecture and security reviews.
- Hiring & people: defined the AI Engineer role, ran technical interviews, made final hiring decisions and negotiated compensation; onboard and mentor engineers.
- Adoption & governance: ran AI enablement sessions for ~600 people across departments (incl. Copilot vs Claude); screened AI requests and analysed Copilot consumption and licensing.

Tech Lead – Data Scientist
Airbus | Portugal | May 2023 – Nov 2025

- Led AI proofs of concept with Global Business Services stakeholders for cross-border material specification matching, copywriting aids and HR support (clustering, text similarity, GenAI).
- Built and deployed GenAI solutions to production on GCP in the Central AI team (RAG, knowledge graphs, agentic workflows) for manufacturing process assistance.
- Industrialised a data anonymisation pipeline on AWS (spaCy, MLflow), improving F1 score by 20–35% across file types and languages.
- Built semantic-similarity data drift detection and clustering of quality non-conformities for predictive maintenance in manufacturing.

Machine Learning & AI Engineer
Nokia | Lisbon, Portugal | Nov 2021 – May 2023

- Developed and deployed NLP models on Azure, improving the global chatbot's answer accuracy by 20 percentage points with text-similarity techniques, benchmarked with QBox.
- Built a career development recommendation system using similarity and clustering methods.

Data Scientist
Accenture | Lisbon, Portugal | Mar 2021 – Nov 2021

- Built predictive models (Python, scikit-learn) for external banking clients to forecast customer response to campaigns; applied clustering and text analysis for audience segmentation and personalised advertising.
- Processed large datasets with Hadoop, NoSQL and AWS, and presented trends and model results to clients in Power BI and Qlik Sense dashboards.

SELECTED TRAINING & CERTIFICATIONS

Advanced Generative AI Google Cloud
Deep Learning Specialization DeepLearning.AI (Coursera)
Machine Learning with scikit-learn Inria

SUMMARY

AI engineering lead and architect delivering ML and GenAI solutions across enterprise software, aerospace, telecommunications and banking. At Unit4, lead a team of 10 consultants (3 direct reports) on 7 production use cases, after shipping 3 document-AI solutions as sole engineer. Hands-on across RAG, agents and document AI on Azure, GCP and AWS.

EDUCATION

MSc Electrical and Computer Engineering
Instituto Superior Técnico | Lisbon, 2021

- Focus: AI/ML, Robotics, Computer Vision

BSc Electrical and Computer Engineering
University of Coimbra | Coimbra, 2018

CORE EXPERTISE & TOOLS

Leadership & Delivery
Stakeholder discovery, solution architecture, team leadership, hiring, build vs buy, AI governance, cost & licensing

AI Engineering
Document AI, RAG, agentic workflows, knowledge graphs, NLP, embeddings, OCR, LLM evaluation

Architecture & Infrastructure
Solution architecture, infrastructure as code (Bicep), Docker, Microsoft Azure, GCP, AWS, REST APIs

Evaluation & Observability
Langfuse, Prometheus, Grafana, MLflow

Programming & Frameworks
Python, SQL, scikit-learn, spaCy, LangChain, LightRAG, PyTorch, TensorFlow

Enterprise Integrations
CRM systems, SharePoint, ServiceNow, Snowflake, Copilot Studio

Development Tools
Git, Azure DevOps, Claude Code, GitHub Copilot

LANGUAGES

Portuguese (Native) | English (C2, EF SET)

PUBLICATION

3D Estimation of Visual Focus of Attention
IEEE International Conference
MSc thesis research, ISR – Instituto de Sistemas e Robótica